Symmetry Spans and Enforced Gaplessness

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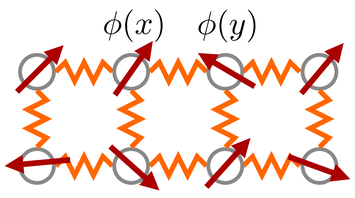
Overview

Bridging Scales

Microscopic model | **Macroscopic phenomena**



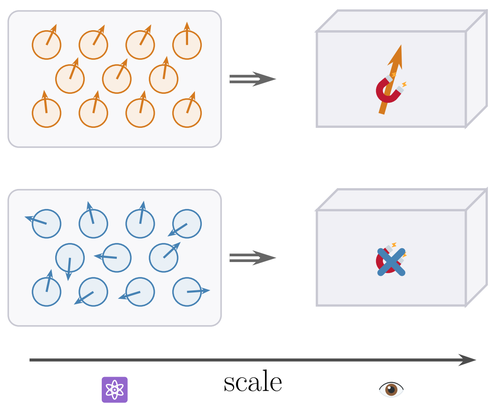
?

gapless (conductor)

gapped (insulator)

Symmetry Classifies Phases

Spins on a lattice with **SO(3) rotation symmetry** can realize different phases:



- **Ferromagnet**: SO(3) spontaneously broken, gapless
- **Paramagnet**: SO(3) preserved, gapped

> ? Can symmetry do more than classify?
> Can it **force** gaplessness — exclude all gapped phases?

Existing Tool: 't Hooft Anomaly Matching

An **anomaly** is a quantum property of a symmetry that persists across scales.

> ! **'t Hooft (1980)**: when a **continuous** symmetry G has an anomaly ($\omega \neq 0$), the IR theory **must be gapless**.

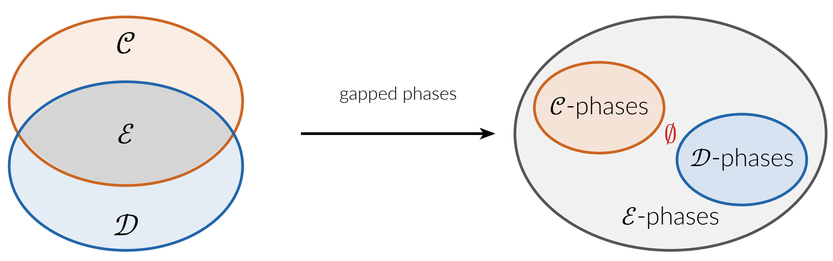
However, perturbative anomalies cannot be realized on a lattice.

> ? Can we find **other** symmetry properties that enforce gaplessness?

New Proposal: Symmetry Span Criterion

A **lattice alternative** to 't Hooft anomaly matching.

Consider **two** symmetries $\mathcal{C}$ and $\mathcal{D}$ acting on a theory, sharing a common sub-symmetry $\mathcal{E}$:



Symmetries | Gapped phases

> ! If no gapped $\mathcal{E}$-phase can be simultaneously a $\mathcal{C}$-phase and a $\mathcal{D}$-phase, then the theory must be **gapless**!

- In all examples: span $\Rightarrow$ continuous symmetry with anomaly $\rightarrow$ **Lattice alternative of continuous 't Hooft anomaly!**
- Applied to lattice spin chains $\rightarrow$ **(right column)**

Technical Details

The Span Setup: 1+1d, continuous G, finite $H \subset G$

$$\begin{tikzcd} \mathrm{Vect}_H \arrow[r, hook, "i_{\mathcal{C}}"] \arrow[d, "i_{\mathcal{D}}"'] & \mathcal{C} \arrow[d, "g"] \\ \mathbf{Vect}_G^{\omega=0} \arrow[r, "f"'] & \mathcal{S}^{\mathrm{full}} \end{tikzcd}$$

- $\mathcal{D} = \mathbf{Vect}_G^{\omega=0}$: continuous symmetry ($G$ connected)
- $\mathcal{C}$: (possibly non-invertible) finite symmetry
- $\mathcal{E} = \mathrm{Vect}_H$: finite subgroup $H \subset G$
- $\mathcal{S}^{\mathrm{full}}$: Full symmetry generated by $\mathcal{C}$ and G.

Span enforcing gaplessness

Gapped phases with symmetry $\mathcal{S} \Leftrightarrow$ TQFTs with $\mathcal{S}$-action.

- $\mathrm{TQFT}(\mathcal{S})$: set of $\mathcal{S}$-symmetric TQFTs.
- $i_{\mathcal{C}}^* : \mathrm{TQFT}(\mathcal{C}) \to \mathrm{TQFT}(\mathcal{E})$: restriction of symmetry

> ! **Criterion:** If $i_{\mathcal{C}}^*\ \mathrm{TQFT}(\mathcal{C}) \cap i_{\mathcal{D}}^*\ \mathrm{TQFT}(G) = \{0\}$, the system admits no symmetric TQFT — it must be gapless.

- $\mathrm{TQFT}(G) \simeq \mathbb{N}[G\text{-SPTs}]$ (no SSB for connected G)

$\mathrm{U}(1) + \mathrm{TY}(\mathbb{Z}_2)$: Kramers–Wannier Duality

- $H = \mathbb{Z}_2, G = \mathrm{U}(1), \mathcal{C} = \mathrm{TY}(\mathbb{Z}_2)$.
- $i_{\mathcal{C}}^*\ \mathrm{TQFT}(\mathrm{TY}(\mathbb{Z}_2)) \ni \left(\mathrm{SSB}_{\mathbb{Z}_2} \oplus \mathrm{triv}_{\mathbb{Z}_2}\right)^n$
  - $\mathrm{SSB}_{\mathbb{Z}_2}$ and $\mathrm{triv}_{\mathbb{Z}_2}$ exchanged by KW duality
- $i_{\mathcal{D}}^*\ \mathrm{TQFT}(\mathrm{U}(1)) \ni \mathrm{triv}_{\mathbb{Z}_2}^m$

> ! Intersection $= \{0\} \Rightarrow$ **no symmetric TQFT!**

Lattice Realizations

$$\begin{tikzcd} \mathrm{Vect}_{\mathbb{Z}_2} \arrow[r, hook, "i_{\mathcal{C}}"] \arrow[d, hook, "i_{\mathcal{D}}"'] & \mathrm{TY}(\mathbb{Z}_2)_{\mathrm{lattice}} \arrow[d] \\ \mathbf{Vect}_{\mathrm{U}(1)}^{\omega=0} \arrow[r] & \mathcal{S}_{\mathrm{XX}} \supset \text{Onsager alg.} \end{tikzcd}$$

XX chain: $\mathcal{H} = \bigotimes_j \mathbb{C}_j^2$, $H = \sum_j (Y_j Z_{j+1} - Z_j Y_{j+1})$

with KW duality $\mathrm{TY}(\mathbb{Z}_2)$ and $\mathrm{U}(1)$: $Q = \sum_j X_j$.

- Span generates **Onsager algebra** [1, 2], $\tilde{Q} = \sum_j Z_j Z_{j+1}$.

$\mathbb{Z}_N$ clock chain: $\mathcal{H} = \bigotimes_j \mathbb{C}_j^N$, $\mathrm{U}(1)$ and $\mathrm{TY}(\mathbb{Z}_N)$.

- Symmetric **interacting** Hamiltonian with Onsager algebra with integrable RG flow: $c = 1, 3(n-1)/(n+1)$ [1].

> ! Span provides a **new lattice argument** for gaplessness:
> $\overset{\mathrm{RG}}{\rightsquigarrow} \mathrm{U}(1)^{\mathrm{shift}} \times \mathrm{U}(1)^{\mathrm{wind}}$ with mixed anomaly

Outlook

- **Higher dimensions:** The span criterion extends naturally; e.g. 3+1d Maxwell theory at special coupling.

> ? **Systematic Hamiltonian search:** Find lattice Hamiltonians preserving gaplessness-enforcing spans
> — *seeking collaborators!*

References

1. Vernier, E., O'Brien, E., Fendley, P.: Onsager symmetries in $U(1)$-invariant clock models. J. Stat. Mech. 1904, 43107 (2019). https://doi.org/10.1088/1742-5468/ab11c0
2. Pace, S.D., Chatterjee, A., Shao, S.-H.: Lattice T-duality from non-invertible symmetries in quantum spin chains. SciPost Phys. 18, 121 (2025). https://doi.org/10.21468/SciPostPhys.18.4.121